The Answers

1. Basic principles of semiconductors

a) Different physical parameters influence the conductivity of a material.

i) the density of carriers and the scattering events (influence velocity/mobility). [2]

ii) it increases [2]

iii) because the carrier density is increasing faster than the scattering events are decreasing the mobility. [4]

Note: Here there was a lack of mention concerning scattering events. This is surprising since most wrote about scattering in i) and below in v)

iv) it decreases [2]

Note: This is A-level physics and draws attention to the difference between a metal (sea of free electrons and semiconductors with only quasi-free electrons)

v) because the carrier density remains approximately constant but the scattering events are increasing thus reducing the velocity of the carriers. [4]

b) Different carrier transport processes determine the currents through a device.

i) electron drift. [2]

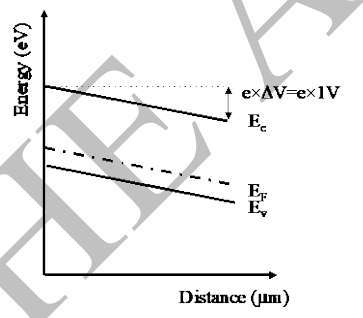
ii) electron diffusion and hole drift in the charge neutral p-type region

hole diffusion and electron drift in the charge neutral n-type region

across the depletion region: diffusion of electrons and drift of holes from n to p region, diffusion of holes and drift of electrons from the p to n region. [6]

Note: Very few students covered all the transport processes for both carriers type in all regions. Marks on this part were very low on average.

c) Energy band diagrams.



i) [4]

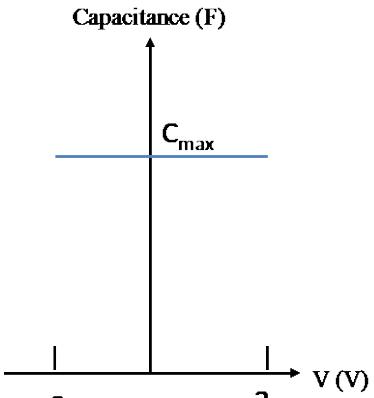
ii) 1 eV [2]

iii) 1 eV. [2]

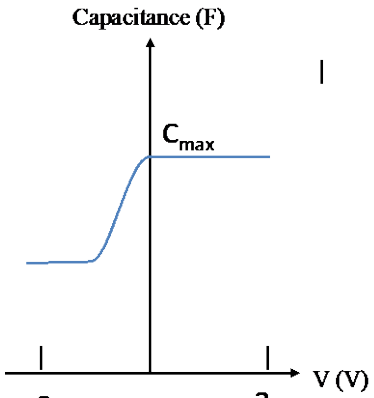
d) $C_{max} = \frac{4\times8.85\times10^{-12}\ \mathrm{F/m}}{100\times10^{-9}m} 100 \times 10^{-12} = 3.54 \times 10^{-14} F$ for both cases

Note: many mistakes again μm^2 -> m^2 conversion. 1 μm = 10^{-6} m thus (1 μm)^2 = (10^{-6} m)^2 = 10^{-12} m^2

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i) [5]



ii) [5]

Note: many mistakes were made against the constant value of a classical parallel plate capacitor. This is A-level material and also seen in Energy conversion and in Circuit Analysis, capacitor values C do not change with voltage. What was most surprising is that there were also many mistaken in ii) due to confusion about the flat band statement that indicates that depletion starts as V=0V. Also quite a few students were not able to even sketch this CV characteristic. Surprising as most were able to sketch the CV of a MOSFET in the next question.

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2. MOSFETs

a) The threshold voltage is the gate voltage that needs to be applied to generate an inversion layer carrier density at the Si/SiO_2 interface of the same magnitude as the carrier density in the bulk. Inversion charge has the opposite charge sign compared to the bulk charge. [4]

Note: this is a definition and this requires precision in the answer. Best way to answer this question is to think that your answer is for someone who has not seen MOSFETs before: what is inversion? What contact does Vth refer to? You must be sure to say something about the density of carriers.

An alternative answer given by a few students is that it is the voltage on the gate to cause the depletion region underneath the gate oxide to be maximum.

b) The energy band diagram from gate through the oxide into the silicon (MOS).

i) p-type. The Fermi level in Si lies closer to valence band than to the conduction band or equivalently the Fermi level lies below the intrinsic level (midgap) in Si. [2]

ii) zero. Because the Fermi levels are constant across the structure. [2]

Note: answer to this question was very disappointing. The majority of the student cohort missed that the Fermi level remains constant and thus the external voltage must be zero. It was especially disappointing as this was extensively covered in the revision lecture. As a consequence of the mistake, iii) was wrongly answered.

iii) $V_{FB} = V_{ox} + V_s = 0.1\ eV + 0.25\ eV = 0.35\ eV$. [4]

iv) The workfunction of the metal is smaller than the workfunction of the Si. Because when the free energy of electrons in the metal is higher than that in Si before contact, the free energy of electrons in the Si must increase upon contact and thus E_c comes closer to E_F as then free electrons can be found at higher energy level. [4]

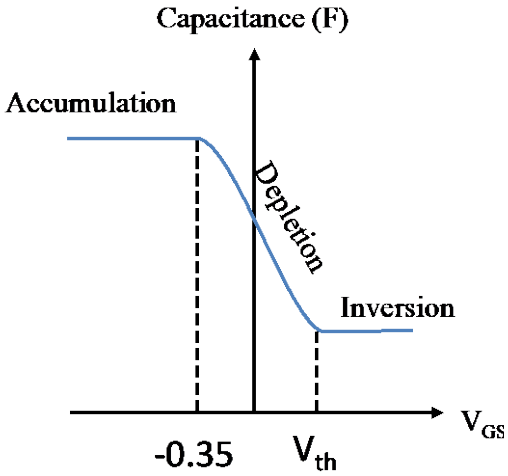
Note: this went well for all students who made a little sketch of the band diagram. That made the thought process really obvious.

v) The threshold voltage will be lower than the case presented in fig. 2.1. For the same initial band bending we need less further bending to ensure the distance between E_c and E_F at the Si/SiO_2 interface is equal to the distance between E_F and E_v in the bulk as E_F-E_v in bulk has increased with lower N_A. [4]

Note: most got this correct by completely ignoring the effect of the original band bending. That's fine as it can be ignored. Some students also plotted an energy band diagram for this that made the answer very clear. My suggestion is: plot an energy band diagram.

c) [10]

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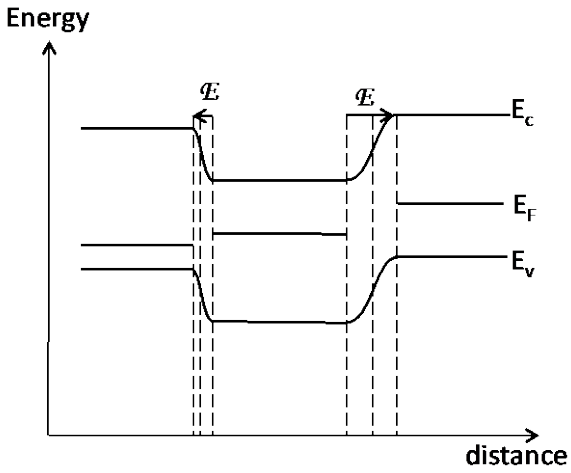
Marks distribution:
Right orientation: 3
Right naming of areas: 3
Shift for accumulation V_{GS}: 2
Values on voltage axis: 2

Note: most students had a reasonable plot for the CV characteristics. Where the voltage point at which accumulation starts (here -0.35V) was consistent and with the right sign to what was answered in biii) no marks were subtracted.

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3. BJT

a) . [10]



Marks distribution:
Correct doping: 3
Correct flat PE: 1
Correct movement of E_F with bias: 4
Correct relative dimensions of depletion widths: 2

b)

i) the emitter current density. [4]

$$J_E = eD_n \frac{n_{p_0} exp\left(V_{EB}/V_T\right) - n_{p_0}}{X_E} + eD_p \frac{p_{n_0} exp\left(V_{EB}/V_T\right) - p_{n_0} exp\left(V_{CB}/V_T\right)}{W_B}$$

$$V_{CB} < 0V \;\&\; exp\left(V_{EB}/V_T\right) \gg 1$$

$$J_E \approx eD_n \frac{n_{p_0} exp\left(V_{EB}/V_T\right)}{X_E} + eD_p \frac{p_{n_0} exp\left(V_{EB}/V_T\right)}{W_B}$$

$$J_E \approx e \times n_i^2 \times exp\left(V_{EB}/V_T\right)\left(\frac{D_n}{N_A X_E} + \frac{D_p}{N_D W_B}\right) =$$
$$1.6\,10^{-19}\,(1.45\,10^{10})^2 exp\left(\frac{0.5}{0.026}\right)\left(\frac{20}{10^{18}\times 10^{-4}} + \frac{12}{10^{16}\times 0.8\times 10^{-4}}\right) = 1.51\times 10^{-3} + 1.13\times 10^{-1} \approx 1.13\times 10^{-1} A/cm^2$$

ii) the collector current density. [4]

$$J_C = eD_n \frac{n_{p_0} exp\left(V_{CB}/V_T\right) - n_{p_0}}{X_C} + eD_p \frac{p_{n_0} exp\left(V_{EB}/V_T\right) - p_{n_0} exp\left(V_{CB}/V_T\right)}{W_B}$$

$$V_{CB} < 0V$$

$$J_C \approx eD_n \frac{-n_{p_0}}{X_C} + eD_p \frac{p_{n_0} exp\left(V_{EB}/V_T\right)}{W_B}$$

$$exp\left(V_{EB}/V_T\right) \gg 1 \;\&\; W_B \ll X_C$$

$$J_C \approx eD_p \frac{p_{n_0} exp\left(V_{EB}/V_T\right)}{W_B} = 1.13\times 10^{-1} A/cm^2$$

iii) the base current density. [2]

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$$J_B = J_E - J_C \approx eD_n \frac{n_{p_0} exp\left(V_{EB}/V_T\right)}{X_E} = 1.51 \times 10^{-3} A/cm^2$$

iv) the current gain. [2]

$\beta = \frac{J_C}{J_B} = \frac{D_p X_E N_A}{D_n W_B N_D} = \frac{1.13\times10^{-1}}{1.51\times10^{-3}} = 75$ with N_A the doping in the emitter and N_D the doping in the base

Note: this question went well in those cases where students plotted the carrier concentrations in all regions and labelled the boundary values with suitable parameters and then took a systematic and ordered approach.

First state that depletion widths are going to be ignored.

Then write the full equation for JE and JC without ignoring any component. Many many errors were made here by stating that the current in the base relates to: p_{n0} (exp(V/V_T) – 1). This is obviously incorrect as it needs to be $p'_n - p''_n = p_{n0}$ (exp(V_{EB}/V_T) – exp(V_{BC}/V_T)). It was also important to state that $|V_{BC}|$=-1V due to the reverse bias across the junction.

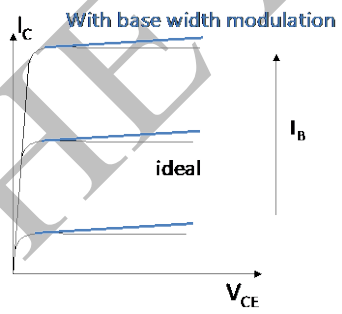
Then make the simplification with justification

Then put in the values. The final result was not important. The correct values for the symbols in the equations are important as that highlights that the correct lengths for the material layers were used. There were a lot of students that used the diffusion lengths (L_n and L_p) whilst the real lengths are given and are clearly shorter than the diffusion length.

Although this question seems like a lot of work, actually after having calculated the values for the simplified equation for IE, the calculations do not have to be repeated. Very few students did that and wasted a lot of time calculating the same multiple times.

c) Output characteristics.

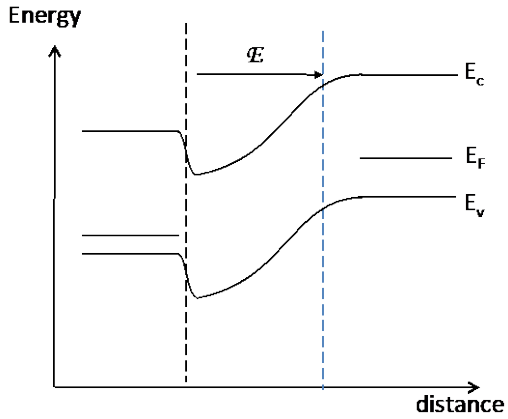
i) When taking the depletion region width that extend into the base into account then the output characteristics show a slope for higher V_{CE} values due to base width modulation – the effective width of the base decreases with increasing reverse bias across the base-collector junction (thus V_{CE} increasing). I_C is inversely proportional to base width. Output conductance $g_0 > 0$ S. [3]



Note: students have seen these plots multiple times in the Analogue Electronics module, thus a surprise when some can't draw the output characteristics. The slope in the "constant current range" is important. At least I-V are expected on the axis V cannot be VBE as that is the input voltage.

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ii) . [3]



iii) When the base is completely depleted then there is a strong electric field and the energy band diagram resembles that of a diode (doped-depletion-doped). We can expect exponential currents when V_{BC} lowers the EB potential barrier for V_{CE} increasing. [2]

Note: these questions were aimed at thinking a bit out of the box and applying basic knowledge of semiconductor devices in a new situation. Well done those who ensured a tilt in the E_c and E_v in the base because it is depleted: electric fields drop across the depletion region. Those who mentioned drift also in iii) got full marks. The precise shape of the bands in the base is not important as long as it is clear that there is a tilt and thus an electric field.

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